Laboration #2 (Examination)

Write a program in any language (example c, java or python etc.) that finds and show the position of the Lidar sensor inside a box. This is about a how to use the Lidar as a sensor, estimate the position and the uncertainty of the estimate. The examination will go like this:

The program must be used and tested with the Raspberry Pi. The program should print the position and the uncertainty of the Lidar sensor inside a box within a console window. This should be done with approximately 1s sample time. Position X,Y values in mm and angle in degree. Uncertainty σ_x, σ_y and σ_θ also in mm and degree. The supervisor will then check the response of the system by relative slowly move the Lidar inside the box a certain distance.

Before the examination you can chose the box. It can be a paper box, the “arena” or something else.

Good Luck!